Final Exam Review

Cumulative: Lessons 1–18, Emphasis on Lessons 10–19

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Today's Agenda

- Exam logistics and structure
- First-half foundations (what's on the exam from Lessons 1–9)
- Second-half concepts — the main focus:
  - Data exploration
  - Paired t-test
  - ANOVA and post-hoc testing
  - Categorical data analysis
  - Nonparametric tests
  - Correlation and simple linear regression
  - Power, sample size, and synthesis
- Choosing the right test
- Your questions

The final exam opens TODAY at 3:00 PM and closes Friday, March 20 at 5:00 PM

Exam Logistics

Exam Details

Format:

- 8 parts, 150 points total
- Designed for 4–6 hours completion
- Mix of conceptual questions, R code, and interpretation

Resources allowed:

- ✅ Course materials (slides, homework, textbook)
- ✅ R documentation and help files (`?t.test`, `?pnorm`, etc.)
- ✅ Your own notes
- ❌ Other students
- ❌ AI assistants (ChatGPT, Claude, etc.)
- ❌ Online forums or external help

Submit both `.qmd` and `.html` files — naming: `Lastname_FirstInitial_Final`

What's Covered

Part 1: Foundations — first half (15 pts)

Part 2: Exploring the data (15 pts)

Part 3: Paired t-test (20 pts)

Part 4: ANOVA and post-hoc (20 pts)

Part 5: Categorical data (20 pts)

Part 6: Nonparametric tests (15 pts)

Part 7: Correlation and SLR (25 pts)

Part 8: Power, sample size, synthesis (20 pts)

Key dataset: `diabetes_trial.csv` — simulated RCT, 180 patients, 3 treatment arms (Metformin, Lifestyle, Combined), 6-month follow-up

General Tips

Before you start:

- Set up your workspace: load packages, read in data
- Make sure code chunks evaluate (`eval: TRUE` or remove `eval: FALSE`)
- Check that `here("data", "diabetes_trial.csv")` finds your file

While working:

- Interpret everything in complete sentences — reference the numbers
- Use all 6 steps for hypothesis tests
- Use comments in your R code to show your thinking
- Render frequently — don't wait until the end

Before submitting:

- Render one final time — confirm no red errors
- Verify both `.qmd` and `.html` are submitted
- Check file naming format

The #1 mistake

Running code that *works* but never rendering the document. A file that won't knit cannot be graded. Render early and often.

Part 1: First-Half Foundations

What from the first half is on the exam?

Part 1 (15 points) tests three targeted areas from Lessons 1–9:

1.1 — Study design (5 pts)

- Experiment vs. observational study
- Why randomization matters
- When can we claim causation?

1.2 — Bayes' theorem / PPV (5 pts)

- Given sensitivity, specificity, prevalence → calculate PPV
- Interpret PPV in context

1.3 — Normal distributions and CLT (5 pts)

- `pnorm()` for an individual value
- `pnorm()` for a sample mean (using SE = $\sigma/\sqrt{n}$)

Tip

Your **midterm review slides** cover all of this in detail — a quick re-read is the best preparation for Part 1.

Study design: the essentials

Experiment:

- Researchers **assign** treatments (random assignment)
- Controls for confounding
- Can establish **causation**

Observational study:

- Researchers observe without interfering
- Cannot control for all confounders
- Can only show **association**

Causation checklist

A study can claim causation **only if:**

1. It is a randomized experiment
2. Random assignment was used
3. Other explanations have been ruled out

The diabetes trial: **experiment** (patients randomized to 3 arms) → causation is potentially claimable ✓

Bayes' Theorem: PPV setup

Given sensitivity, specificity, and prevalence:

$$PPV = P(\text{Disease} \mid \text{Test}+) = \frac{\text{sensitivity} \times \text{prevalence}}{\text{sensitivity} \times \text{prevalence} + (1 - \text{specificity}) \times (1 - \text{prevalence})}$$

```r
sensitivity <- 0.82
specificity <- 0.90
prevalence  <- 0.25

ppv <- (sensitivity * prevalence) /
       (sensitivity * prevalence + (1 - specificity) * (1 - prevalence))
ppv
```

Tip

Interpreting PPV: "Of all patients who test positive, ___% actually have the condition."

A high PPV → test is useful for **confirming** disease. A low PPV → test is better for **screening** (ruling out).

Normal distributions and CLT: two scenarios

Individual observation: $X \sim N(\mu, \sigma^2)$

```r
# P(X > 110) if X ~ N(95, 12)
pnorm(110, mean = 95, sd = 12,
      lower.tail = FALSE)
```

The SD here is σ — variability of **individuals**

Sample mean: $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$

```r
# P(Xbar > 100) for n=36, mu=95, sigma=12
se <- 12 / sqrt(36)
pnorm(100, mean = 95, sd = se,
      lower.tail = FALSE)
```

The SD here is $SE = \sigma/\sqrt{n}$ — variability of **sample means**

The key distinction

When the question says "a random individual" → use σ.

When the question says "a sample of n" → use $SE = \sigma/\sqrt{n}$.

Part 2: Exploring the Data

Descriptive statistics by group

```r
# Overall summary
diabetes %>%
  summarise(
    n    = n(),
    mean = mean(hba1c_baseline, na.rm = TRUE),
    sd   = sd(hba1c_baseline, na.rm = TRUE),
    median = median(hba1c_baseline, na.rm = TRUE)
  )

# By group
diabetes %>%
  group_by(treatment) %>%
  summarise(
    n    = n(),
    mean = mean(hba1c_change, na.rm = TRUE),
    sd   = sd(hba1c_change, na.rm = TRUE)
  )

# Or with rstatix:
diabetes %>%
  group_by(treatment) %>%
  get_summary_stats(hba1c_change, type = "mean_sd")
```

Visualization: histograms and boxplots

```r
# Histogram — check shape, skewness, outliers
ggplot(diabetes, aes(x = triglycerides)) +
  geom_histogram(bins = 30) +
  labs(x = "Triglycerides (mg/dL)", y = "Count",
       title = "Distribution of triglycerides")

# Boxplots by group with jitter
ggplot(diabetes, aes(x = treatment, y = hba1c_6mo, fill = treatment)) +
  geom_boxplot(alpha = 0.6) +
  geom_jitter(width = 0.2, alpha = 0.4) +
  labs(x = "Treatment", y = "HbA1c at 6 months (%)",
       title = "HbA1c at 6 months by treatment group")
```

Tip

`geom_jitter()` is layered **after** `geom_boxplot()` so points appear on top. Always include `width = 0.2` or similar to prevent overplotting.

Frequency tables with `tabyl()`

```r
# One-way table
diabetes %>%
  tabyl(achieved_target)

# Two-way table with row percentages
diabetes %>%
  tabyl(treatment, achieved_target) %>%
  adorn_totals(where = "row") %>%
  adorn_percentages("row") %>%
  adorn_pct_formatting(digits = 1) %>%
  adorn_ns()
```

Tip

The exam asks for **row percentages** — what % of each treatment group achieved the target. `adorn_percentages("row")` gives you that. Use `adorn_ns()` to keep the raw counts alongside.

Part 3: Paired t-test

When is a paired design appropriate?

Paired (dependent) design:

- Same subject measured **twice** (before/after)
- Matched pairs (sibling controls, eye studies)
- The pairing controls for between-subject variability

In the exam: HbA1c measured at baseline and at 6 months for **the same patient** → naturally paired

Why pairing matters

If you ignore the pairing and treat the two measurements as independent groups, you **throw away information** about within-subject change. The paired test uses differences $d_i = Y_{i,\text{after}} - Y_{i,\text{before}}$ and is equivalent to a one-sample t-test on those differences.

Ignoring pairing → larger SE → less power → more likely to miss a real effect.

Paired t-test: the workflow

Step 1: Create the difference variable

```r
# Positive values = improvement (baseline > 6mo)
diabetes <- diabetes %>%
  mutate(hba1c_change = hba1c_baseline - hba1c_6mo)
```

Step 2: Run the test

```r
# Option A: one-sample t-test on differences
t.test(diabetes$hba1c_change, mu = 0, alternative = "two.sided")

# Option B: paired t-test directly
t.test(diabetes$hba1c_baseline, diabetes$hba1c_6mo,
       paired = TRUE, alternative = "two.sided")
```

Tip

Both give identical results. Option A is clearer for applying the six steps — you are testing whether the mean difference equals zero.

Paired t-test: the six steps

Step 1: $H_0 : \mu_d = 0$ (no mean change) vs. $H_A : \mu_d \neq 0$ (or > 0 if directional)

Step 2: $\alpha = 0.05$

Step 3: Assumptions

- Independence: one measure per patient ✓
- Normality of differences: $n = 180$ is large → CLT applies ✓

Step 4: $t = \dfrac{\bar{d} - 0}{s_d/\sqrt{n}}$ — reported as `t` in `t.test()` output

Step 5: p-value from `t.test()` output, $df = n - 1 = 179$

Step 6: If p < 0.05 → reject H_0. Report in context:

> "There was a statistically significant mean decrease in HbA1c from baseline to 6 months (t = ***, df = 179, p < 0.001). The mean decrease was*** percentage points (95% CI: ___ to ___)."

Part 4: ANOVA and Post-Hoc Testing

When to use one-way ANOVA

ANOVA: comparing means across 3+ groups

Use one-way ANOVA when:

- **Outcome:** continuous
- **Predictor:** categorical with **3 or more groups**
- **Question:** Do any of the group means differ?

In the exam: Does mean HbA1c change (`hba1c_change`) differ across Metformin, Lifestyle, and Combined groups?

Why not just run three t-tests?

Each t-test has a 5% Type I error rate. With 3 pairwise comparisons, the family-wise error rate inflates to ~14% — you're much more likely to find a spurious significant result.

ANOVA assumptions

Check before running:

1. **Independence:** one observation per patient (study design) ✓
2. **Normality:** approximately normal within each group — check with QQ plot or note large n
3. **Equal variance:** SD ratio (max SD / min SD) should be < 2

```r
# Check group-level n, mean, SD
diabetes %>%
  group_by(treatment) %>%
  summarise(
    n  = n(),
    mean = mean(hba1c_change, na.rm = TRUE),
    sd   = sd(hba1c_change, na.rm = TRUE)
  ) %>%
  mutate(sd_ratio = max(sd) / min(sd))
```

Tip

SD ratio < 2 → equal variance assumption is met.

ANOVA: the six steps + R code

Step 1: $H_0 : \mu_{\text{Met}} = \mu_{\text{Life}} = \mu_{\text{Comb}}$ vs. H_A : at least one group mean differs

Steps 2–3: $\alpha = 0.05$; check assumptions ✓

Steps 4–5: Fit and summarize

```r
anova_fit <- aov(hba1c_change ~ treatment, data = diabetes)

# Traditional output
summary(anova_fit)

# Tidy output
broom::tidy(anova_fit)
```

Step 6: Report F-statistic with both df values and p-value

> "There was a significant effect of treatment on HbA1c change (F(2, 177) = **, *p* =**)."

Post-hoc testing: Tukey's HSD

If ANOVA rejects H_0, use Tukey HSD to identify **which pairs** differ:

```
# Tukey HSD
TukeyHSD(anova_fit)
```

Reading the output:

- `diff` = difference in group means
- `p adj` = adjusted p-value (controls family-wise error rate)
- Report pairs where `p adj < 0.05`

Writing a results paragraph (Q 4.4)

Include: descriptive stats for each group + F-statistic + p-value + specific post-hoc comparisons.

> "Mean HbA1c change was ___ (SD = ***) in the Metformin group,*** (SD = ***) in the Lifestyle group, and*** (SD = ***) in the Combined group. A one-way ANOVA indicated a significant difference across groups (F(2, 177) =*** , p = ___). Tukey post-hoc tests revealed that [Combined vs. Metformin: mean diff = ***, p =***] differed significantly, while [other comparisons] did not."

Part 5: Categorical Data Analysis

Chi-squared test for independence

When to use: Both variables are **categorical** and you have 3+ groups

In the exam: Is `treatment` (3 groups) associated with `achieved_target` (Yes/No)?

```r
# Step 1: Build the contingency table
tab <- diabetes %>%
  tabyl(treatment, achieved_target)

# Step 2: Check expected counts
chisq.test(tab)$expected   # All should be ≥ 5

# Step 3: Run the test
chisq.test(tab)
```

Hypotheses (in words):

- H_0: There is no association between treatment and achieving the glycemic target
- H_A: There is an association between treatment and achieving the glycemic target

Two-proportion z-test

When to use: Two groups, binary outcome — testing $H_0 : p_1 = p_2$

In the exam: Combined vs. Lifestyle groups only — who achieved the target at a higher rate?

```r
# Filter to two groups
diabetes_two <- diabetes %>%
  filter(treatment %in% c("Combined", "Lifestyle"))

# Compute successes and group sizes
counts <- diabetes_two %>%
  group_by(treatment) %>%
  summarise(
    success = sum(achieved_target == "Yes"),
    n = n()
  )

# Two-proportion test
prop.test(x = counts$success,
          n = counts$n,
          alternative = "two.sided")
```

Chi-squared vs. two-proportion test

	Chi-squared	Two-proportion (`prop.test`)
Groups	2 or more	Exactly 2
Outcome	Categorical (any # levels)	Binary (success/failure)
Hypotheses	Independence vs. association	$p_1 = p_2$ vs. $p_1 \neq p_2$
Output	χ^2, df, p-value	z (or χ^2), p-value, **CI for difference**

Tip

For exactly **2 groups and a binary outcome**, both tests give equivalent p-values. Prefer `prop.test()` when you want a **confidence interval for the difference in proportions** — which gives more information than the p-value alone.

Part 6: Nonparametric Tests

When to go nonparametric

The decision is about the data, not just the sample size

Use a nonparametric test when:

- The distribution is **heavily skewed** or has extreme outliers
- The outcome is **ordinal** (pain scales, severity scores)
- Sample sizes are small **and** normality is implausible

In the exam: `triglycerides` — visualized as right-skewed in Part 2 → Kruskal-Wallis is appropriate

SD ratio check: large SD ratio across groups also supports nonparametric approach

Warning

Small n alone does not automatically mean nonparametric. If the data look approximately normal, the parametric test is valid and more powerful. The criterion is whether the assumptions are plausibly met.

Three nonparametric tests: the map

Parametric test	Nonparametric alternative	R function
One-sample / paired t-test	Wilcoxon signed-rank	`wilcox.test(x, mu = 0)`
Two-sample t-test	Wilcoxon rank-sum (Mann-Whitney)	`wilcox.test(x, y)`
One-way ANOVA	Kruskal-Wallis	`kruskal_test()`

Hypotheses (state correctly!):

- H_0: The distributions of [outcome] are identical across groups
- H_A: At least one group has a different distribution

Important

Do **not** state hypotheses in terms of medians being equal — this is only valid under additional assumptions. "Distributions are identical" is the correct framing.

Kruskal-Wallis + pairwise follow-up

```r
# Kruskal-Wallis (nonparametric ANOVA)
diabetes %>%
  kruskal_test(triglycerides ~ treatment)

# Pairwise Wilcoxon with FDR correction (if K-W is significant)
diabetes %>%
  wilcox_test(triglycerides ~ treatment,
              p.adjust.method = "fdr") %>%
  filter(p.adj < 0.05)
```

Tip

Use **FDR** (Benjamini-Hochberg) for multiple comparisons in the nonparametric setting — it controls the **false discovery rate** and is less conservative than Bonferroni when you have several comparisons.

Part 7: Correlation and Simple Linear Regression

This section is 25 points — the largest on the exam

No homework covered this material

Part 7 is the only section that was not practiced in a homework assignment. The exam provides extra guidance (hints in code chunks). Read the questions carefully and follow the steps.

The material maps directly to Lesson 21 slides — a review of those slides before the exam is time well spent.

What Part 7 tests:

- Scatterplot + regression line
- Pearson and Spearman correlation: `cor()`
- Fitting `lm()`, interpreting `tidy()` output
- LINE assumption checks with `augment()` + diagnostic plots
- R^2 from `glance()`
- Prediction vs. confidence interval: `predict()`

Scatterplot and correlation

```r
# Scatterplot with regression line
ggplot(diabetes, aes(x = bmi, y = fasting_glucose)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(x = "BMI (kg/m²)", y = "Fasting glucose (mg/dL)",
       title = "Fasting glucose vs. BMI")

# Pearson correlation: strength of LINEAR association
cor(diabetes$bmi, diabetes$fasting_glucose,
    use = "complete.obs")

# Spearman: rank-based, robust to skew/outliers
cor(diabetes$bmi, diabetes$fasting_glucose,
    method = "spearman",
    use = "complete.obs")
```

Describing the relationship: direction (positive/negative), form (linear/curved), strength (weak/moderate/strong), unusual observations

Fitting the regression model

```r
# Fit
glucose_model <- lm(fasting_glucose ~ bmi, data = diabetes)

# Coefficients, SE, t, p, CI
tidy(glucose_model, conf.int = TRUE)

# Model-level summary: R², sigma, n
glance(glucose_model)
```

The estimated equation:

$$\widehat{\text{Fasting Glucose}} = \hat{\beta}_0 + \hat{\beta}_1 \times \text{BMI}$$

Interpreting the slope ($\hat{\beta}_1$):

> "For every 1 kg/m² increase in BMI, fasting glucose is expected to increase by $\hat{\beta}_1$ mg/dL, on average."

Interpreting the intercept ($\hat{\beta}_0$): Often not meaningful — only interpret when $X = 0$ is realistic.

The LINE conditions

Letter	Condition	How to check
L	**L**inearity — relationship is linear	Scatterplot; residual plot
I	**I**ndependence — observations independent	Study design
N	**N**ormality — residuals approximately normal	QQ plot of residuals
E	**E**qual variance — residuals have constant spread	Residual plot

```r
aug <- augment(glucose_model)

# Residual plot (checks L and E)
ggplot(aug, aes(x = .fitted, y = .resid)) +
  geom_point(alpha = 0.4) +
  geom_hline(yintercept = 0, linetype = "dashed") +
  labs(x = "Fitted values", y = "Residuals", title = "Residual plot")

# QQ plot (checks N)
ggplot(aug, aes(sample = .resid)) +
  stat_qq() + stat_qq_line() +
  labs(title = "QQ plot of residuals")
```

Reading the diagnostic plots

Residual plot (L and E)

Good: Points scatter randomly around 0; flat smoother; roughly equal spread

Red flags:

- Curve → linearity violated (L)
- Funnel → equal variance violated (E)

QQ plot (N)

Good: Points track the diagonal reference line

Red flags:

- S-shape → skewed residuals
- Heavy tails → points deviate at both ends

Note: With $n = 180$, the CLT makes moderate departures from normality less of a concern. L and E are more critical for inference.

R^2 and sigma

```r
glance(glucose_model) %>%
  select(r.squared, sigma, nobs)
```

R^2 interpretation

"BMI explains ___% of the variability in fasting glucose in this sample."

R^2 ranges from 0 (model explains nothing) to 1 (perfect fit).

For SLR: $R^2 = r^2$ (square of Pearson correlation)

sigma

`sigma` = residual standard deviation = typical distance of an individual observation from the fitted line, in units of Y.

This is the individual-level scatter — much larger than the `std.error` for the slope, and it's why prediction intervals are so wide.

Inference for the slope: six steps

Step 1: $H_0 : \beta_1 = 0$ (BMI tells us nothing about glucose) vs. $H_A : \beta_1 \neq 0$

Step 2: $\alpha = 0.05$

Step 3: Check LINE conditions ✓

Step 4: $t = \hat{\beta}_1 / SE_{\hat{\beta}_1}$ — read from `tidy()` output, `statistic` column

Step 5: p-value from `tidy()` output, `p.value` column; $df = n - 2$

Step 6: Conclusion in context:

> "There is [strong/insufficient] evidence of a positive linear association between BMI and fasting glucose (t = ***, df =*** , p = ***). For every 1 kg/m² increase in BMI, fasting glucose is expected to increase by*** mg/dL (95% CI: ___ to ___)."

CI for mean response vs. prediction interval

	Question	Accounts for
CI for mean response	Average glucose for **all patients** with BMI = 30?	Uncertainty about the line
Prediction interval	Glucose for **this specific new patient** with BMI = 30?	Uncertainty about the line **+** individual variation ($\hat{\sigma}$)

```r
new_data <- data.frame(bmi = c(25, 30, 35))

# CI for mean response
predict(glucose_model, newdata = new_data, interval = "confidence")

# Prediction interval for a new individual
predict(glucose_model, newdata = new_data, interval = "prediction")
```

Important

Prediction intervals are **always wider** than CIs for mean response. More data narrows the CI (better estimated line) but cannot eliminate individual variation around the line.

Part 8: Power, Sample Size, and Synthesis

Power analysis with `pwr.t.test()`

Cohen's d: effect size for comparing two means

$$d = \frac{\mu_1 - \mu_2}{\sigma_{\text{pooled}}}$$

```r
# How large a sample do we need?
pwr.t.test(n         = NULL,
           d         = 0.4 / 0.50,   # delta / pooled SD
           sig.level = 0.05,
           power     = 0.80,
           type      = "two.sample",
           alternative = "two.sided")

# What power do we have with n = 40 per group?
pwr.t.test(n         = 40,
           d         = 0.4 / 0.50,
           sig.level = 0.05,
           power     = NULL,
           type      = "two.sample",
           alternative = "two.sided")
```

Tip

Always **round up** the required n — you can't have a fractional person, and rounding down means you're slightly underpowered.

Sample size and what it affects

When you increase n:

	Effect
ANOVA F-statistic	Increases (more power to detect group differences)
Post-hoc p-values	Decrease (more power per pairwise comparison)
Width of CIs	Narrower (smaller SE → tighter interval)

Peeking at data before choosing the test

Running ANOVA, seeing which group "won," then doing a t-test comparing that group vs. the others is **data-driven test selection** — it inflates the Type I error rate and violates the pre-specified analysis plan. This is the issue raised in Question 8.2b. The right approach: specify your comparisons **before** looking at the data.

Choosing the right test: the synthesis table

Research question	Groups	Outcome	Test	Key R function
Did HbA1c change? (same patient)	1 (differences)	Continuous	Paired t-test	`t.test(d, mu = 0)`
Does mean change differ across 3 treatments?	3	Continuous, normal	One-way ANOVA	`aov()`
Do triglycerides differ across 3 groups? (skewed)	3	Continuous, skewed	Kruskal-Wallis	`kruskal_test()`
Is treatment associated with achieving target? (3 groups)	3	Binary	Chi-squared	`chisq.test()`
Does target rate differ between 2 groups?	2	Binary	Two-proportion z-test	`prop.test()`
Is there a linear association between BMI and glucose?	—	2 continuous	SLR / slope test	`lm()` + `tidy()`

CI and hypothesis test: two sides of the same coin

Question 8.3b asks you to connect these

For a two-sided test at $\alpha = 0.05$:

- If the **95% CI excludes** the null value → p-value < 0.05 → reject H_0
- If the **95% CI includes** the null value → p-value ≥ 0.05 → fail to reject H_0

What the CI adds beyond the p-value:

- The **direction and magnitude** of the effect (not just "significant or not")
- A range of **plausible values** for the true population parameter
- Clinical interpretability — a 0.5 mmHg drug effect may be statistically significant but clinically meaningless

Good example to use for this question: The paired t-test CI for mean HbA1c change, or the CI for the regression slope — both tell you not just *whether* the effect is real, but *how big* it plausibly is.

Reminders and Good Luck

Common mistakes to avoid

1. **Ignoring paired structure** — running an independent t-test on before/after data
2. **Forgetting to check assumptions** — especially ANOVA's SD ratio and chi-squared expected counts
3. **Misframing nonparametric hypotheses** — say "distributions are identical," not "medians are equal"
4. **Confusing σ, s, SE, and `sigma` from `glance()`** — know what each measures
5. **Using `lower.tail = TRUE/FALSE` incorrectly** — sketch the picture first
6. **Not rendering before submitting** — code that works in a chunk but breaks the render is ungraded

The most important R check before submitting

Remove or set `eval: TRUE` on any chunks you may have set to `eval: FALSE` while working. A chunk that doesn't evaluate won't produce output, and you'll lose points.

You are ready

- ✅ You have worked through 8 homework assignments with increasingly complex analyses
- ✅ You have practiced the six-step framework across many test types
- ✅ You have hands-on experience with every R function on this exam
- ✅ Part 7 has hints — you are not going in cold on SLR

The exam opens at 3:00 PM today. You have until Friday at 5:00 PM.

Read carefully, interpret in context, render frequently, and trust your preparation.

Good luck!

Questions?

Your Turn

What would you like to go over?

- Concepts that feel unclear?
- R functions you want to see again?
- Specific parts of the exam structure?

This is your time — ask anything.